Раздел 1

1. Создайте на сервере для 1 практики ключ ssh при помощи программы ssh-keygen

   Скопируйте созданный ключ на сервер для 2 практики для пользователя root при помощи программы ssh-copy-id

```
eltex-pg2-v21@eltex-2026-spring:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/eltex-pg2-v21/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/eltex-pg2-v21/.ssh/id_rsa
Your public key has been saved in /home/eltex-pg2-v21/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:m7BsKbcT6qaT3ZDpuuqPKKKORXCqrvsLYkL84WBSnH0 eltex-pg2-v21@eltex-2026-spring
The key's randomart image is:
+---[RSA 3072]----+
|                 |
| . o             |
|. = . E          |
|.=   .           |
|o=.. o. S        |
|=.+ =..+ o       |
|=..*.+*.o        |
|Xo+ =+o.         |
|#OBX. ..         |
+----[SHA256]-----+
eltex-pg2-v21@eltex-2026-spring:~$ ssh-copy-id root@172.16.9.221
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/eltex-pg2-v21/.ssh/id_rsa.pub"
The authenticity of host '172.16.9.221 (172.16.9.221)' can't be established.
ED25519 key fingerprint is SHA256:YNVzj+RtetZfXv5XUn/CUjeEcaMD4eg3jy8EuGGIS+o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new key
root@172.16.9.221's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@172.16.9.221'"
and check to make sure that only the key(s) you wanted were added.
```

2. Подключитесь к серверу для 2 практики под пользователем root и сравните содержимое файла открытого ключа на сервере 1 ~/.ssh/*.pub и файла ~/.ssh/authorized_keys на сервере для 2 практики, а так же права доступа для каждого из файлов.

```
eltex-pg2-v21@eltex-2026-spring:~$ ssh root172.16.9.221
ssh: Could not resolve hostname root172.16.9.221: Name or service not known
eltex-pg2-v21@eltex-2026-spring:~$ ssh root@172.16.9.221
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-55-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Mar 14 06:30:42 PM UTC 2026

  System load:  0.0                Processes:              111
  Usage of /:   41.1% of 14.66GB   Users logged in:        0
  Memory usage: 10%                IPv4 address for ens18: 172.16.9.221
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

87 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


*** System restart required ***
Last login: Sat Mar 14 17:55:41 2026 from 172.16.8.4
```

```
eltex-pg2-v21@eltex-2026-spring:~$ cat ~/.ssh/*.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDEC2hRUvH4P47owfZEO/eu
a2zpwah3tKvX0rkNbSphu3Ief/f4XwUuFRwzhuY7FI7LhONk/ixA4pdG7Qd/
wLanp9svv7hZ5/aaOE1FMNNwrS9MFmXV2ZP6Fk0v5D/bq0ps/1a/tBc4B3BW
UE1QHNfwPXDE9/EWSISWsSzMCkVWOiv8dovhkcehfWFcqqXQr9LJhHMo/Ler
AHecwL57DyB1Kg4c34WeDhnT3H70149X9ure7YzcSBO2CuU1YAERwd8bq4Hn
T/Wd8bgwjZZH7dr8w0ptgWYnZ+kPzCiiJw+X6V4wKYgFoKfsen0gJjiPSXmZ
PO4avvie0Qc7oskkDKSNWHSdjPCLccKqFAdrNkNiRmykw1LSLqO2ybFEE75O
OW1s1943GfEBk2H7M73MYyBVHyiMI613mlGnhlRopRLtFyOqIY5oP1TXW5+H
O+9a149Z0k5nGticOmxe0OQy1VZcb5SUc/MIO1ZVglk6B13H0mj6FuNg1U7d
BrK2weHC7P8= eltex-pg2-v21@eltex-2026-spring
eltex-pg2-v21@eltex-2026-spring:~$
eltex-pg2-v21@eltex-2026-spring:~$
eltex-pg2-v21@eltex-2026-spring:~$
eltex-pg2-v21@eltex-2026-spring:~$ ls -la ~/.ssh/*.pub
-rw-r--r-- 1 eltex-pg2-v21 eltex-pg2-v21 585 Mar 15 01:29 /h
ome/eltex-pg2-v21/.ssh/id_rsa.pub
eltex-pg2-v21@eltex-2026-spring:~$
```

```
Last login: Sat Mar 14 17:55:41 2026 from 172.16.8.4
root@eltex-practice2-pg2-v21:~# cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDEC2hRUvH4P47owfZEO/eua
2zpwah3tKvX0rkNbSphu3Ief/f4XwUuFRwzhuY7FI7LhONk/ixA4pdG7Qd/wL
anp9svv7hZ5/aaOE1FMNNwrS9MFmXV2ZP6Fk0v5D/bq0ps/1a/tBc4B3BWUE1
QHNfwPXDE9/EWSISWsSzMCkVWOiv8dovhkcehfWFcqqXQr9LJhHMo/LerAHec
wL57DyB1Kg4c34WeDhnT3H70149X9ure7YzcSBO2CuU1YAERwd8bq4HnT/Wd8
bgwjZZH7dr8w0ptgWYnZ+kPzCiiJw+X6V4wKYgFoKfsen0gJjiPSXmZPO4avv
ie0Qc7oskkDKSNWHSdjPCLccKqFAdrNkNiRmykw1LSLqO2ybFEE75OOW1s194
3GfEBk2H7M73MYyBVHyiMI613mlGnhlRopRLtFyOqIY5oP1TXW5+HO+9a149Z
0k5nGticOmxe0OQy1VZcb5SUc/MIO1ZVglk6B13H0mj6FuNg1U7dBrK2weHC7
P8= eltex-pg2-v21@eltex-2026-spring
root@eltex-practice2-pg2-v21:~#
root@eltex-practice2-pg2-v21:~#
root@eltex-practice2-pg2-v21:~#
root@eltex-practice2-pg2-v21:~# ls -la ~/.ssh/authorized_keys
-rw------- 1 root root 585 Mar 14 18:30 /root/.ssh/authorized
_keys
root@eltex-practice2-pg2-v21:~#
```

Содержимое файлов одинаково. На сервере 1 практики кто угодно может читать ключ, так как он публичный. На сервере 2 права только у root.

3. Создайте пользователя user1 при помощи команды useradd, укажите необходимость создания домашнего каталога и shell /bin/bash. Создайте пароль пользователю user1.
   Создайте пользователя user2 и user3 при помощи команды adduser.

```
root@eltex-practice2-pg2-v21:~# useradd -m -s /bin/bash user1
root@eltex-practice2-pg2-v21:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
root@eltex-practice2-pg2-v21:~#
```

```
root@eltex-practice2-pg2-v21:~# useradd -m -s /bin/bash user1
root@eltex-practice2-pg2-v21:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
root@eltex-practice2-pg2-v21:~# adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1002) ...
info: Adding new user `user2' (1002) with group `user2 (1002)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `user2' to supplemental / extra groups `users' ...
info: Adding user `user2' to group `users' ...
root@eltex-practice2-pg2-v21:~#
```

4. Для пользователя user3 смените shell на /usr/sbin/nologin (man usermod), выполните вход под этим пользователем при помощи утилиты su, сначала без дополнительных параметров, затем с явным указанием shell /bin/bash в параметрах su. Выполните logout

```
root@eltex-practice2-pg2-v21:~# usermod -s /usr/sbin/nologin user3
root@eltex-practice2-pg2-v21:~# su user3
This account is currently not available.
root@eltex-practice2-pg2-v21:~# su --shell /bin/bash user3
user3@eltex-practice2-pg2-v21:/root$ logout
bash: logout: not login shell: use `exit'
user3@eltex-practice2-pg2-v21:/root$ exit
exit
root@eltex-practice2-pg2-v21:~#
```

5. Создайте новую группу и добавьте её для всех пользователей user* как дополнительную, посмотрите список групп всех пользователей user*

```
root@eltex-practice2-pg2-v21:~# usermod -a -G group user1
root@eltex-practice2-pg2-v21:~# usermod -a -G group user2
root@eltex-practice2-pg2-v21:~# usermod -a -G group user3
root@eltex-practice2-pg2-v21:~# groups user1 user2 user3
user1 : user1 group
user2 : user2 users group
user3 : user3 users group
root@eltex-practice2-pg2-v21:~#
```

6. Создайте каталог /opt/share и назначьте группу из предыдущего пункта его владельцем, установите на этот каталог бит SGID, права для группы rwx.

```
root@eltex-practice2-pg2-v21:~# mkdir -p /opt/share
root@eltex-practice2-pg2-v21:~#
root@eltex-practice2-pg2-v21:~# chown :group /opt/share
root@eltex-practice2-pg2-v21:~# chmod 2770 /opt/share
root@eltex-practice2-pg2-v21:~#
```

7. Для user1 задайте перманентно umask, снимающий право чтения для «прочих»

   Echo “umask 0006” >> /home/user1/.bashrc

```
root@eltex-practice2-pg2-v21:~# echo "umask 0006" >>  /home/user1/.bashrc
root@eltex-practice2-pg2-v21:~#
```

```
Last login: Sat Mar 14 19:42:10 2026 from 172.16.8.4
root@eltex-practice2-pg2-v21:~# su user1
user1@eltex-practice2-pg2-v21:/root$ umask
0006
user1@eltex-practice2-pg2-v21:/root$
```

8. Создайте каждым из пользователей новые файлы в каталоге /opt/share, удалите файлы созданные другими пользователями.

```
root@eltex-practice2-pg2-v21:~# su user3
This account is currently not available.
root@eltex-practice2-pg2-v21:~# su --shell /bin/bash user3
user3@eltex-practice2-pg2-v21:/root$ touch /opt/share/user3.txt
user3@eltex-practice2-pg2-v21:/root$ ls -l /opt/share
total 0
-rw-rw---- 1 user1 group 0 Mar 15 06:38 user1.txt
-rw-rw-r-- 1 user2 group 0 Mar 15 06:38 user2.txt
-rw-rw-r-- 1 user3 group 0 Mar 15 06:41 user3.txt
user3@eltex-practice2-pg2-v21:/root$ rm /opt/share/user1.txt
user3@eltex-practice2-pg2-v21:/root$ ls -l
ls: cannot open directory '.': Permission denied
user3@eltex-practice2-pg2-v21:/root$ ls -l /opt/share
total 0
-rw-rw-r-- 1 user2 group 0 Mar 15 06:38 user2.txt
-rw-rw-r-- 1 user3 group 0 Mar 15 06:41 user3.txt
user3@eltex-practice2-pg2-v21:/root$ su user2
Password:
user2@eltex-practice2-pg2-v21:/root$ rm /opt/share/user3.txt
user2@eltex-practice2-pg2-v21:/root$ su user1
Password:
1234su: Authentication failure
user2@eltex-practice2-pg2-v21:/root$ su user1
Password:
user1@eltex-practice2-pg2-v21:/root$ rm /opt/share/user2.txt
user1@eltex-practice2-pg2-v21:/root$ ls -l /opt/share
total 0
user1@eltex-practice2-pg2-v21:/root$
```

   Файлы созданные другими пользователем без проблем удаляются.

9. Повторите предыдущий пункт, предварительно установив sticky bit на каталоге /opt/share

   chmod +t /opt/share

```
root@eltex-practice2-pg2-v21:~# su --shell /bin/bash user3
user3@eltex-practice2-pg2-v21:/root$ touch /opt/share/user3.txt
user3@eltex-practice2-pg2-v21:/root$ ls -l /opt/share
total 0
-rw-rw---- 1 user1 group 0 Mar 15 06:50 user1.txt
-rw-rw-r-- 1 user2 group 0 Mar 15 06:50 user2.txt
-rw-rw-r-- 1 user3 group 0 Mar 15 06:52 user3.txt
user3@eltex-practice2-pg2-v21:/root$ rm /opt/share/user2.txt
rm: cannot remove '/opt/share/user2.txt': Operation not permitted
user3@eltex-practice2-pg2-v21:/root$ su user2
Password:
user2@eltex-practice2-pg2-v21:/root$ rm /opt/share/user1.txt
rm: cannot remove '/opt/share/user1.txt': Operation not permitted
user2@eltex-practice2-pg2-v21:/root$ su user1
Password:
user1@eltex-practice2-pg2-v21:/root$ rm /opt/share/user3.txt
rm: cannot remove '/opt/share/user3.txt': Operation not permitted
user1@eltex-practice2-pg2-v21:/root$
```

Теперь только владелец файла или root может удалить файл (или переименовать).

10. Разрешите user1 выполнять привилегированную команду dmesg при помощи команды sudo, а user2 – при помощи скрипта на языке bash с установленным флагом SUID.

    echo "user1 ALL=(ALL) /bin/dmesg" >> /etc/sudoers
    touch /home/user2/dmesg_user2.sh
    chmod +x /home/user2/dmesg_user2.sh
    chmod u+s /home/user2/dmesg_user2.sh

```
root@eltex-practice2-pg2-v21:~# chmod +x /home/user2/dmesg_user2.sh
root@eltex-practice2-pg2-v21:~# chmod u+s /home/user2/dmesg_user2.sh
root@eltex-practice2-pg2-v21:~# ls -l /home/user2
total 0
-rwsr-xr-x 1 root root 0 Mar 15 07:09 dmesg_user2.sh
-rw-r--r-- 1 root root 0 Mar 15 07:09 dmesg_user2.shtouch
root@eltex-practice2-pg2-v21:~#
```

```
root@eltex-practice2-pg2-v21:/home/user2# su user2
user2@eltex-practice2-pg2-v21:~$ ./dmesg_user2.sh
dmesg: read kernel buffer failed: Operation not permitted
user2@eltex-practice2-pg2-v21:~$ ls -l
total 4
-rwsr-xr-x 1 root root 24 Mar 15 07:23 dmesg_user2.sh
user2@eltex-practice2-pg2-v21:~$
```

    dmesg_user2.sh:
    #!/bin/bash
    /bin/dmesg

    Когда мы запускаем bash скрипт, система /bin/bash и передаёт ему файл. Bash при этом сбрасывает привилегии до уровня user2, поэтому dmesg выдаёт ошибку доступа.

```
user1@eltex-practice2-pg2-v21:/home/user2$ sudo dmesg | head
[sudo] password for user1:
[    0.000000] Linux version 6.8.0-55-generic (buildd@lcy02-amd64-095) (x86_64-linux-gnu-gcc-13 (Ubuntu
 13.3.0-6ubuntu2~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #57-Ubuntu SMP PREEMPT_DYNAMIC W
ed Feb 12 23:42:21 UTC 2025 (Ubuntu 6.8.0-55.57-generic 6.8.12)
[    0.000000] Command line: BOOT_IMAGE=/vmlinuz-6.8.0-55-generic root=/dev/mapper/ubuntu--vg-ubuntu--l
v ro
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
[    0.000000]   Hygon HygonGenuine
[    0.000000]   Centaur CentaurHauls
[    0.000000]   zhaoxin   Shanghai
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
user1@eltex-practice2-pg2-v21:/home/user2$
```

11. Для всех пользователей user* задайте время действия пароля – 10 дней.

    Chage -M 10 user1

```
root@eltex-practice2-pg2-v21:~# chage -M 10 user1
root@eltex-practice2-pg2-v21:~# chage -M 10 user2
root@eltex-practice2-pg2-v21:~# chage -M 10 user3
root@eltex-practice2-pg2-v21:~#
```

12. Отредактируйте файл /etc/motd, вписав туда свое имя и фамилию.

    echo "Чиркин Артём" >> /etc/motd

```
 System information as of Sun Mar 15 08:47:52 AM UTC 2026

  System load:  0.12               Processes:              109
  Usage of /:   41.2% of 14.66GB   Users logged in:        0
  Memory usage: 11%                IPv4 address for ens18: 172.16.9.221
  Swap usage:   0%

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   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

87 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


*** System restart required ***
Чиркин Артём
Last login: Sun Mar 15 06:37:16 2026 from 172.16.8.4
root@eltex-practice2-pg2-v21:~#
```

13. Создайте копию содержимого каталога /etc в каталог /root/etc_backup при помощи программы rcync.

    rsync -a /etc/ /root/etc_backup/

```
root@eltex-practice2-pg2-v21:~# rsync -a /etc/ /root/etc_backup
root@eltex-practice2-pg2-v21:~# ls /root
etc_backup  opt
root@eltex-practice2-pg2-v21:~#
```

14. Заархивируйте содержимое каталога /root/etc_backup архиватором tar, используйте алгоритмы сжатия gzip, bzip2, 7zip, сравните размеры полученных файлов.

    tar -czf /root/etc_backup.tar.gz /root/etc_backup
    tar -cjf /root/etc_backup.tar.bz2 /root/etc_backup
    tar -cf - /root/etc_backup | 7za a -si /root/etc_backup.7z

```
root@eltex-practice2-pg2-v21:~# ls -lh /root/etc_backup.*
-rw-r--r-- 1 root root 476K Mar 15 09:20 /root/etc_backup.7z
-rw-r--r-- 1 root root 554K Mar 15 09:17 /root/etc_backup.tar.bz2
-rw-r--r-- 1 root root 584K Mar 15 09:13 /root/etc_backup.tar.gz
root@eltex-practice2-pg2-v21:~#
```

    7zip сжал лучше всех.

15. Отредактируйте файл /etc/motd, вписав туда текущую дату и время, синхронизируйте каталог /root/etc_backup с каталогом /etc при помощи rsync, добавьте файл motd в архив, сжатый gzip.

    echo "Current date: $(date)" >> /etc/motd
    rsync -av /etc/ /root/etc_backup

```
root@eltex-practice2-pg2-v21:~# echo "Current date: $(date)" >> /etc/motd
root@eltex-practice2-pg2-v21:~# rsync -av /etc/ /root/etc_backup
sending incremental file list
motd

sent 70,018 bytes  received 283 bytes  140,602.00 bytes/sec
total size is 2,366,893  speedup is 33.67
root@eltex-practice2-pg2-v21:~#
```

    gunzip etc_backup.tar.gz
    tar -rf /root/etc_backup.tar /root/etc_backup/motd
    gzip etc_backup.tar

```
root@eltex-practice2-pg2-v21:~#
root@eltex-practice2-pg2-v21:~# gunzip etc_backup.tar.gz
root@eltex-practice2-pg2-v21:~# ls
etc_backup  etc_backup.7z  etc_backup.tar  etc_backup.tar.bz2  opt
root@eltex-practice2-pg2-v21:~# tar -rf /root/etc_backup.tar /root/etc_backup/motd
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
root@eltex-practice2-pg2-v21:~# gzip etc_backup.tar
root@eltex-practice2-pg2-v21:~# ls
etc_backup  etc_backup.7z  etc_backup.tar.bz2  etc_backup.tar.gz  opt
root@eltex-practice2-pg2-v21:~#
```

16. Сравните содержимое архива, упакованного bzip2 с сорержимым каталога /root/etc_backup.

mkdir /tmp/archive_extract
tar -xjf etc_backup.tar.bz2 -C /tmp/archive_extract
diff -r /tmp/archive_extract /root/etc_backup
rm -rf /tmp/archive_extract

```
1 Socket in /run/screen/S-eltex-pg2-v21.
eltex-pg2-v21@eltex-2026-spring:~$ screen -dr
Only in /root/etc_backup: passwd
Only in /root/etc_backup: passwd-
Only in /root/etc_backup: perl
Only in /root/etc_backup: pki
Only in /root/etc_backup: plymouth
Only in /root/etc_backup: pm
Only in /root/etc_backup: polkit-1
Only in /root/etc_backup: pollinate
Only in /root/etc_backup: profile
Only in /root/etc_backup: profile.d
Only in /root/etc_backup: protocols
Only in /root/etc_backup: .pwd.lock
Only in /root/etc_backup: python3
Only in /root/etc_backup: python3.12
Only in /root/etc_backup: qemu
Only in /root/etc_backup: rc0.d
Only in /root/etc_backup: rc1.d
Only in /root/etc_backup: rc2.d
Only in /root/etc_backup: rc3.d
Only in /root/etc_backup: rc4.d
Only in /root/etc_backup: rc5.d
Only in /root/etc_backup: rc6.d
Only in /root/etc_backup: rcS.d
Only in /root/etc_backup: resolv.conf
Only in /root/etc_backup: .resolv.conf.systemd-resolved.bak
Only in /root/etc_backup: rmt
Only in /tmp/archive_extract: root
Only in /root/etc_backup: rpc
Only in /root/etc_backup: rsyslog.conf
Only in /root/etc_backup: rsyslog.d
Only in /root/etc_backup: screenrc
Only in /root/etc_backup: security
Only in /root/etc_backup: selinux
Only in /root/etc_backup: sensors3.conf
Only in /root/etc_backup: sensors.d
Only in /root/etc_backup: services
Only in /root/etc_backup: sgml
Only in /root/etc_backup: shadow
Only in /root/etc_backup: shadow-
Only in /root/etc_backup: shells
Only in /root/etc_backup: skel
Only in /root/etc_backup: sos
Only in /root/etc_backup: ssh
Only in /root/etc_backup: ssl
Only in /root/etc_backup: subgid
Only in /root/etc_backup: subgid-
Only in /root/etc_backup: subuid
Only in /root/etc_backup: subuid-
Only in /root/etc_backup: sudo.conf
Only in /root/etc_backup: sudoers
Only in /root/etc_backup: sudoers.d
Only in /root/etc_backup: sudo_logsrvd.conf
Only in /root/etc_backup: supercat
Only in /root/etc_backup: sysctl.conf
Only in /root/etc_backup: sysctl.d
Only in /root/etc_backup: sysstat
Only in /root/etc_backup: systemd
Only in /root/etc_backup: terminfo
Only in /root/etc_backup: thermald
Only in /root/etc_backup: timezone
Only in /root/etc_backup: tmpfiles.d
Only in /root/etc_backup: ubuntu-advantage
Only in /root/etc_backup: ucf.conf
Only in /root/etc_backup: udev
Only in /root/etc_backup: udisks2
Only in /root/etc_backup: ufw
```

17. Распакуйте архивы etc_backup, упакованные gzip и 7zip в каталоги /root/etc_backup_gzip и /root/etc_backup_7zip, сравните программой diff файлы motd в этих каталогах.

    mkdir etc_backup_gzip
    tar -xzf etc_backup.tar.gz -C /root/etc_backup_gzip

```
root@eltex-practice2-pg2-v21:~# mkdir etc_backup_gzip
root@eltex-practice2-pg2-v21:~# tar -xzf etc_backup.tar.gz -C /root/etc_backup_gzip
```

    mkdir etc_backup_7zip
    7z x etc_backup.7z -o/root/etc_backup_7zip
    cd etc_backup_7zip
    tar -xf etc_backup

    diff etc_backup_gzip/root/etc_backup/motd
    etc_backup_7zip/root/etc_backup/motd

```
< Current date: Sun Mar 15 09:27:39 AM UTC 2026
root@eltex-practice2-pg2-v21:~#
root@eltex-practice2-pg2-v21:~# ls -la etc_backup_gzip/root/etc_backup/motd
-rw-r--r-- 1 root root 70 Mar 15 09:27 etc_backup_gzip/root/etc_backup/motd
root@eltex-practice2-pg2-v21:~# ls -la etc_backup_7zip/root/etc_backup/motd
-rw-r--r-- 1 root root 24 Mar 15 08:47 etc_backup_7zip/root/etc_backup/motd
root@eltex-practice2-pg2-v21:~# diff etc_backup_gzip/root/etc_backup/motd etc_backup_7zip/root/etc_backup/mo>
2d1
< Current date: Sun Mar 15 09:27:39 AM UTC 2026
root@eltex-practice2-pg2-v21:~#
```

    Файлы идентичны, кроме даты, которую мы сами вписали.

Раздел 2

1. Найдите все записи из лога загрузки, доступного через команду journalctl с опцией -b в первые 20 секунд с момента загрузки, начало загрузки определить автоматически.

    sudo journalctl -b --until "+20s"

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Mar 16 00:12:36 2026 from 212.164.65.4
eltex-pg2-v21@eltex-2026-spring:~$ screen -dr
Mar 12 15:58:28 eltex-2026-spring systemd[3907450]: Reached target basic.target - Basic System.
Mar 12 15:58:28 eltex-2026-spring systemd[3907450]: Reached target default.target - Main User Target.
Mar 12 15:58:28 eltex-2026-spring systemd[3907450]: Startup finished in 76ms.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Activating special unit exit.target...
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target default.target - Main User Target.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target basic.target - Basic System.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target paths.target - Paths.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target sockets.target - Sockets.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target timers.target - Timers.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Closed dbus.socket - D-Bus User Message Bus Socket.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Removed slice app.slice - User Application Slice.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Reached target shutdown.target - Shutdown.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Finished systemd-exit.service - Exit the Session.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Reached target exit.target - Exit the Session.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Queued start job for default target default.target.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Created slice app.slice - User Application Slice.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target paths.target - Paths.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target timers.target - Timers.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Starting dbus.socket - D-Bus User Message Bus Socket...
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Listening on dbus.socket - D-Bus User Message Bus Socket
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target sockets.target - Sockets.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target basic.target - Basic System.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target default.target - Main User Target.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Startup finished in 79ms.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Activating special unit exit.target...
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target default.target - Main User Target.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target basic.target - Basic System.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target paths.target - Paths.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target sockets.target - Sockets.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target timers.target - Timers.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Closed dbus.socket - D-Bus User Message Bus Socket.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Removed slice app.slice - User Application Slice.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Reached target shutdown.target - Shutdown.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Finished systemd-exit.service - Exit the Session.
```

```
Mar 12 15:58:28 eltex-2026-spring systemd[3907450]: Reached target basic.target - Basic System.
Mar 12 15:58:28 eltex-2026-spring systemd[3907450]: Reached target default.target - Main User Target.
Mar 12 15:58:28 eltex-2026-spring systemd[3907450]: Startup finished in 76ms.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Activating special unit exit.target...
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target default.target - Main User Target.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target basic.target - Basic System.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target paths.target - Paths.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target sockets.target - Sockets.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Stopped target timers.target - Timers.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Closed dbus.socket - D-Bus User Message Bus Socket.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Removed slice app.slice - User Application Slice.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Reached target shutdown.target - Shutdown.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Finished systemd-exit.service - Exit the Session.
Mar 12 15:59:10 eltex-2026-spring systemd[3907450]: Reached target exit.target - Exit the Session.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Queued start job for default target default.target.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Created slice app.slice - User Application Slice.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target paths.target - Paths.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target timers.target - Timers.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Starting dbus.socket - D-Bus User Message Bus Socket...
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target sockets.target - Sockets.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target basic.target - Basic System.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Reached target default.target - Main User Target.
Mar 12 17:09:33 eltex-2026-spring systemd[4012991]: Startup finished in 79ms.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Activating special unit exit.target...
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target default.target - Main User Target.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target basic.target - Basic System.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target paths.target - Paths.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target sockets.target - Sockets.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Stopped target timers.target - Timers.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Closed dbus.socket - D-Bus User Message Bus Socket.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Removed slice app.slice - User Application Slice.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Reached target shutdown.target - Shutdown.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Finished systemd-exit.service - Exit the Session.
Mar 12 18:48:27 eltex-2026-spring systemd[4012991]: Reached target exit.target - Exit the Session.
Mar 13 01:03:40 eltex-2026-spring systemd[552237]: Queued start job for default target default.target.
Mar 13 01:03:40 eltex-2026-spring systemd[552237]: Created slice app.slice - User Application Slice.
Mar 13 01:03:40 eltex-2026-spring systemd[552237]: Reached target paths.target - Paths.
Mar 13 01:03:40 eltex-2026-spring systemd[552237]: Reached target timers.target - Timers.
Mar 13 01:03:40 eltex-2026-spring systemd[552237]: Starting dbus.socket - D-Bus User Message Bus Socket...
Mar 13 01:03:41 eltex-2026-spring systemd[552237]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Mar 13 01:03:41 eltex-2026-spring systemd[552237]: Reached target sockets.target - Sockets.
Mar 13 01:03:41 eltex-2026-spring systemd[552237]: Reached target basic.target - Basic System.
Mar 13 01:03:41 eltex-2026-spring systemd[552237]: Reached target default.target - Main User Target.
Mar 13 01:03:41 eltex-2026-spring systemd[552237]: Startup finished in 90ms.
eltex-pg2-v21@eltex-2026-spring:~$ clear
```

2. Используя awk найдите все источники и их сообщения в файле auth.log (найдите его find), в названии источника удалите информацию об идентификаторе процесса при помощи sed, полученный результат отсортируйте по названию источника.

   find /.. -name "auth.log"

```
root@eltex-practice2-pg2-v21:~# find /.. -name "auth.log"
/../var/log/auth.log
```

cat /var/log/auth.log | awk '{src=$3;$1=$2=$3=""; print src ": " $0}' | sed -E 's/\[[0-9]+\]+[:]//' | sort

```
eltex-pg2-v21@eltex-2026-spring:~$ screen -dr
su:    pam_unix(su:session): session closed for user user2
su:    pam_unix(su:session): session closed for user user2
su:    pam_unix(su:session): session closed for user user2
su:    pam_unix(su:session): session closed for user user2
su:    pam_unix(su:session): session closed for user user2
su:    pam_unix(su:session): session closed for user user2
su:    pam_unix(su:session): session closed for user user3
su:    pam_unix(su:session): session closed for user user3
su:    pam_unix(su:session): session closed for user user3
su:    pam_unix(su:session): session closed for user user3
su:    pam_unix(su:session): session opened for user user1(uid=1001) by root(uid=0)
su:    pam_unix(su:session): session opened for user user1(uid=1001) by root(uid=0)
su:    pam_unix(su:session): session opened for user user1(uid=1001) by root(uid=1002)
su:    pam_unix(su:session): session opened for user user1(uid=1001) by root(uid=1002)
su:    pam_unix(su:session): session opened for user user1(uid=1001) by root(uid=1002)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=0)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=0)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=0)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=0)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=0)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=1001)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=1003)
su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=1003)
su:    pam_unix(su:session): session opened for user user3(uid=1003) by root(uid=0)
su:    pam_unix(su:session): session opened for user user3(uid=1003) by root(uid=0)
su:    pam_unix(su:session): session opened for user user3(uid=1003) by root(uid=0)
su:    pam_unix(su:session): session opened for user user3(uid=1003) by root(uid=1002)
su:    (to user1) root on pts/0
su:    (to user1) root on pts/0
su:    (to user1) user2 on pts/0
su:    (to user1) user2 on pts/0
su:    (to user1) user2 on pts/0
su:    (to user2) root on pts/0
su:    (to user2) root on pts/0
su:    (to user2) root on pts/0
su:    (to user2) root on pts/0
su:    (to user2) root on pts/0
su:    (to user2) user1 on pts/0
su:    (to user2) user3 on pts/0
su:    (to user2) user3 on pts/0
su:    (to user3) root on pts/0
su:    (to user3) root on pts/0
su:    (to user3) root on pts/0
su:    (to user3) user2 on pts/0
systemd-logind:    New session 523 of user root.
systemd-logind:    New session 541 of user root.
systemd-logind:    New session 553 of user root.
```

```
systemd-logind:    New session 523 of user root.
systemd-logind:    New session 541 of user root.
systemd-logind:    New session 553 of user root.
systemd-logind:    New session 601 of user root.
systemd-logind:    New session 603 of user root.
systemd-logind:    Removed session 523.
systemd-logind:    Removed session 541.
systemd-logind:    Removed session 553.
systemd-logind:    Removed session 601.
systemd-logind:    Session 523 logged out. Waiting for processes to exit.
systemd-logind:    Session 541 logged out. Waiting for processes to exit.
systemd-logind:    Session 553 logged out. Waiting for processes to exit.
systemd-logind:    Session 601 logged out. Waiting for processes to exit.
(systemd)::    pam_unix(systemd-user:session): session opened for user root(uid=0) by root(uid=0)
(systemd)::    pam_unix(systemd-user:session): session opened for user root(uid=0) by root(uid=0)
root@eltex-practice2-pg2-v21:~#
```

3. Для результата из предыдущего пункта найдите количество повторений для каждого источника и выведите их в виде списка «число_повторений источник», результат отсортируйте по убыванию количества повторений

cat /var/log/auth.log | awk '{src=$3;$1=$2=$3=""; print src ": " $0}' | sed -E 's/\[[0-9]+\]+[:]//' | sort | uniq -c | sort -rn

```
 136 CRON:    pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
 136 CRON:    pam_unix(cron:session): session closed for user root
   8 su:    pam_unix(su:session): session closed for user user2
   7 sudo::    pam_unix(sudo:session): session closed for user root
   6 sudo::    pam_unix(sudo:session): session opened for user root(uid=0) by root(uid=0)
   5 su:    (to user2) root on pts/0
   5 su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=0)
   5 su:    pam_unix(su:session): session closed for user user1
   5 sshd:    pam_unix(sshd:session): session opened for user root(uid=0) by root(uid=0)
   4 su:    pam_unix(su:session): session closed for user user3
   4 sshd:    pam_unix(sshd:session): session closed for user root
   3 su:    (to user3) root on pts/0
   3 su:    (to user1) user2 on pts/0
   3 su:    pam_unix(su:session): session opened for user user3(uid=1003) by root(uid=0)
   3 su:    pam_unix(su:session): session opened for user user1(uid=1001) by root(uid=1002)
   2 (systemd)::    pam_unix(systemd-user:session): session opened for user root(uid=0) by root(uid=0)
   2 su:    (to user2) user3 on pts/0
   2 su:    (to user1) root on pts/0
   2 su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=1003)
   2 su:    pam_unix(su:session): session opened for user user1(uid=1001) by root(uid=0)
   2 nologin::    Attempted login by root (UID: 1003) on /dev/pts/0
   1 systemd-logind:    Session 601 logged out. Waiting for processes to exit.
   1 systemd-logind:    Session 553 logged out. Waiting for processes to exit.
   1 systemd-logind:    Session 541 logged out. Waiting for processes to exit.
   1 systemd-logind:    Session 523 logged out. Waiting for processes to exit.
   1 systemd-logind:    Removed session 601.
   1 systemd-logind:    Removed session 553.
   1 systemd-logind:    Removed session 541.
   1 systemd-logind:    Removed session 523.
   1 systemd-logind:    New session 603 of user root.
   1 systemd-logind:    New session 601 of user root.
   1 systemd-logind:    New session 553 of user root.
   1 systemd-logind:    New session 541 of user root.
   1 systemd-logind:    New session 523 of user root.
   1 su:    (to user3) user2 on pts/0
   1 su:    (to user2) user1 on pts/0
   1 su:    pam_unix(su:session): session opened for user user3(uid=1003) by root(uid=1002)
   1 su:    pam_unix(su:session): session opened for user user2(uid=1002) by root(uid=1001)
   1 su::    pam_unix(su:auth): authentication failure; logname=root uid=1002 euid=0 tty=/dev/pts/0 ruser=user2 rhost= user=user1
   1 su:    FAILED SU (to user1) user2 on pts/0
   1 sudo::    user1 : TTY=pts/0 ; PWD=/home/user2 ; USER=root ; COMMAND=/bin/dmesg
   1 sudo::    root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/journalctl --since '2026-03-12 08:28:13 UTC' --until '2026-03-12 08:28:13 UTC' +
sec
   1 sudo::    root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/journalctl --since '2026-03-12 08:28:13 UTC' --until '2026-03-12 08:28:13 UTC +2
sec'
   1 sudo::    root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/journalctl --since '2026-03-12 08:28:13 UTC'
   1 sudo::    root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/journalctl --since 2026-03-12 08:28:13 --until 2026-03-12 08:28:13 + 20 sec
   1 sudo::    root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/journalctl --since '2026-03-12 08:28:13' --until '2026-03-12 08:28:13 +20 sec'
   1 sudo::    root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/journalctl --since '2026-03-12 08:28:13' --until '2026-03-12 08:28:13 + 20 sec'
   1 sudo::    pam_unix(sudo:session): session opened for user root(uid=0) by root(uid=1001)
   1 sshd:    Received disconnect from 172.16.8.4 port 59278:11: disconnected by user
   1 sshd:    Received disconnect from 172.16.8.4 port 58612:11: disconnected by user
   1 sshd:    Received disconnect from 172.16.8.4 port 44496:11: disconnected by user
   1 sshd:    Received disconnect from 172.16.8.4 port 41906:11: disconnected by user
   1 sshd:    Disconnected from user root 172.16.8.4 port 59278
   1 sshd:    Disconnected from user root 172.16.8.4 port 58612
   1 sshd:    Disconnected from user root 172.16.8.4 port 44496
   1 sshd:    Disconnected from user root 172.16.8.4 port 41906
   1 sshd:    Accepted publickey for root from 172.16.8.4 port 59278 ssh2: RSA SHA256:m7BsKbcT6qaT3ZDpuuqPKKKORXCqrvsLYkL84WBSnH0
   1 sshd:    Accepted publickey for root from 172.16.8.4 port 58612 ssh2: RSA SHA256:m7BsKbcT6qaT3ZDpuuqPKKKORXCqrvsLYkL84WBSnH0
   1 sshd:    Accepted publickey for root from 172.16.8.4 port 52652 ssh2: RSA SHA256:m7BsKbcT6qaT3ZDpuuqPKKKORXCqrvsLYkL84WBSnH0
   1 sshd:    Accepted publickey for root from 172.16.8.4 port 44496 ssh2: RSA SHA256:m7BsKbcT6qaT3ZDpuuqPKKKORXCqrvsLYkL84WBSnH0
   1 sshd:    Accepted publickey for root from 172.16.8.4 port 41906 ssh2: RSA SHA256:m7BsKbcT6qaT3ZDpuuqPKKKORXCqrvsLYkL84WBSnH0
   1 chage:    changed password expiry for user3
   1 chage:    changed password expiry for user2
   1 chage:    changed password expiry for user1
```

4. В файле /etc/passwd найдите всех пользователей в системе, у которых установлен shell /usr/sbin/nologin и выведите их в виде списка: «UID, username, список его групп», отсортированный в обратном порядке по UID, список групп используйте из файла /etc/group.

```
#!/bin/bash

while IFS=: read -r user _ uid gid _ _ shell; do
  if [[ "$shell" == "/usr/sbin/nologin" ]]; then
    groups=$(id -Gn $user | tr ' ' ',')
    echo "$uid, $user, $groups"
  fi
done < /etc/passwd | sort -rn
```

```
#/bin/bash


while IFS=: read -r user _ uid gid _ _ shell; do
        if [[ "$shell" == "/usr/sbin/nologin" ]]; then
                groups=$(id -Gn $user | tr ' ' ',')
                echo "$uid , $user , $groups"
        fi
done < /etc/passwd | sort -rn
~
```

```
root@eltex-practice2-pg2-v21:~/opt# ls
task4.sh
root@eltex-practice2-pg2-v21:~/opt# chmod +x task4.sh
root@eltex-practice2-pg2-v21:~/opt# ./task4.sh
65534 , nobody , nogroup
1003 , user3 , user3,users,group
998 , systemd-network , systemd-network
997 , systemd-timesync , systemd-timesync
992 , systemd-resolve , systemd-resolve
991 , polkitd , polkitd
989 , fwupd-refresh , fwupd-refresh
109 , sshd , nogroup
108 , usbmux , plugdev
107 , landscape , landscape
105 , tcpdump , tcpdump
104 , uuidd , uuidd
103 , syslog , syslog,adm
101 , messagebus , messagebus
42 , _apt , nogroup
39 , irc , irc
38 , list , list
34 , backup , backup
33 , www-data , www-data
13 , proxy , proxy
10 , uucp , uucp
9 , news , news
8 , mail , mail
7 , lp , lp
6 , man , man
5 , games , games
3 , sys , sys
2 , bin , bin
1 , daemon , daemon
root@eltex-practice2-pg2-v21:~/opt#
```

5. Найдите в результате вывода dmesg все строки, содержащие слово 'kernel‘

   dmesg | grep -i "kernel"

```
root@eltex-practice2-pg2-v21:~/opt# cd ..
root@eltex-practice2-pg2-v21:~# dmesg | grep -i "kernel"
[    0.000000] KERNEL supported cpus:
[    0.106994] Booting paravirtualized kernel on KVM
[    0.108355] Kernel command line: BOOT_IMAGE=/vmlinuz-6.8.0-55-generic root=/dev/mapper/ubuntu--vg-ubuntu--lv ro
[    0.108419] Unknown kernel command line parameters "BOOT_IMAGE=/vmlinuz-6.8.0-55-generic", will be passed to us
er space.
[    0.305405] Memory: 3933488K/4193760K available (22528K kernel code, 4443K rwdata, 14344K rodata, 4988K init, 4
716K bss, 260012K reserved, 0K cma-reserved)
[    0.515913] DMA: preallocated 512 KiB GFP_KERNEL pool for atomic allocations
[    0.515956] DMA: preallocated 512 KiB GFP_KERNEL|GFP_DMA pool for atomic allocations
[    0.515998] DMA: preallocated 512 KiB GFP_KERNEL|GFP_DMA32 pool for atomic allocations
[    1.066443] Loaded X.509 cert 'Build time autogenerated kernel key: e0bedf6fa049a1e09bd65ebab3904f14cdd5bc68'
[    1.086542] Loaded X.509 cert 'Canonical Ltd. Kernel Module Signing: 88f752e560a1e0737e31163a466ad7b70a850c19'
[    1.273797] Loaded X.509 cert 'Build time autogenerated kernel key: e0bedf6fa049a1e09bd65ebab3904f14cdd5bc68'
[    1.331513] Freeing unused kernel image (initmem) memory: 4988K
[    1.332381] Write protecting the kernel read-only data: 38912k
[    1.336445] Freeing unused kernel image (rodata/data gap) memory: 2040K
[    9.734776] systemd[1]: Listening on systemd-udevd-kernel.socket - udev Kernel Socket.
[    9.819135] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File System...
[    9.843551] systemd[1]: Mounting sys-kernel-tracing.mount - Kernel Trace File System...
[   10.067100] systemd[1]: Starting modprobe@configfs.service - Load Kernel Module configfs...
[   10.102613] systemd[1]: Starting modprobe@dm_mod.service - Load Kernel Module dm_mod...
[   10.116692] systemd[1]: Starting modprobe@drm.service - Load Kernel Module drm...
[   10.143690] systemd[1]: Starting modprobe@efi_pstore.service - Load Kernel Module efi_pstore...
[   10.178821] systemd[1]: Starting modprobe@fuse.service - Load Kernel Module fuse...
[   10.203630] systemd[1]: Starting modprobe@loop.service - Load Kernel Module loop...
[   10.275716] systemd[1]: Starting systemd-modules-load.service - Load Kernel Modules...
[   10.296983] systemd[1]: Starting systemd-remount-fs.service - Remount Root and Kernel File Systems...
root@eltex-practice2-pg2-v21:~#
```

6. Подсчитайте количество строк в файле /var/log/kern.log
   wc -l /var/log/kern.log

```
root@eltex-practice2-pg2-v21:~# wc -l /var/log/kern.log
10 /var/log/kern.log
root@eltex-practice2-pg2-v21:~#
```

7. Отформатируйте вывод записей в /var/log/apt/history.log в следующем порядке, построчно:
   Commandline: ... ; Start-Date: ... ; End-Date: ...

   awk '
   /Start-Date:/ {start=$0}
   /Commandline:/ {cmd=$0}
   /End-Date:/ {print cmd " ; " start " ; " $0}
   ' /var/log/apt/history.log

```
Commandline: apt-get --quiet --assume-yes --option=Dpkg::options::=--force-unsafe-io --option=Dpkg::Options::=--force-confold install linux-g
eneric ; Start-Date: 2025-03-12  18:08:37 ; End-Date: 2025-03-12  18:11:58
Commandline: apt-get --quiet --assume-yes --option=Dpkg::options::=--force-unsafe-io --option=Dpkg::Options::=--force-confold install openssh
-server ; Start-Date: 2025-03-12  18:12:45 ; End-Date: 2025-03-12  18:13:11
Commandline: /usr/bin/unattended-upgrades -v ; Start-Date: 2025-03-12  18:13:21 ; End-Date: 2025-03-12  18:14:11
Commandline: apt dist-upgrade ; Start-Date: 2025-03-12  18:19:08 ; End-Date: 2025-03-12  18:21:04
Commandline: apt install vim qemu-guest-agent ; Start-Date: 2025-03-12  18:21:23 ; End-Date: 2025-03-12  18:21:31
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:00:31 ; End-Date: 2026-03-13  07:06:10
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:06:38 ; End-Date: 2026-03-13  07:11:25
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:11:31 ; End-Date: 2026-03-13  07:18:26
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:18:36 ; End-Date: 2026-03-13  07:20:58
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:21:01 ; End-Date: 2026-03-13  07:23:44
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:23:47 ; End-Date: 2026-03-13  07:24:47
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:24:50 ; End-Date: 2026-03-13  07:27:12
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:27:16 ; End-Date: 2026-03-13  07:29:54
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:30:05 ; End-Date: 2026-03-13  07:31:27
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:31:39 ; End-Date: 2026-03-13  07:38:51
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:39:05 ; End-Date: 2026-03-13  07:41:20
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:41:28 ; End-Date: 2026-03-13  07:45:11
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:45:17 ; End-Date: 2026-03-13  07:47:02
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:47:16 ; End-Date: 2026-03-13  07:48:20
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  07:48:24 ; End-Date: 2026-03-13  08:01:24
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:01:40 ; End-Date: 2026-03-13  08:02:34
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:02:43 ; End-Date: 2026-03-13  08:07:46
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:07:48 ; End-Date: 2026-03-13  08:13:13
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:13:18 ; End-Date: 2026-03-13  08:20:13
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:20:16 ; End-Date: 2026-03-13  08:36:04
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:36:08 ; End-Date: 2026-03-13  08:38:15
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:38:30 ; End-Date: 2026-03-13  08:42:22
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:42:26 ; End-Date: 2026-03-13  08:45:21
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:45:27 ; End-Date: 2026-03-13  08:49:13
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:49:26 ; End-Date: 2026-03-13  08:52:42
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:52:47 ; End-Date: 2026-03-13  08:53:31
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:53:37 ; End-Date: 2026-03-13  08:54:42
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:54:55 ; End-Date: 2026-03-13  08:55:46
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:55:50 ; End-Date: 2026-03-13  08:56:42
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:56:48 ; End-Date: 2026-03-13  08:59:46
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  08:59:51 ; End-Date: 2026-03-13  09:02:10
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:02:15 ; End-Date: 2026-03-13  09:05:59
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:06:03 ; End-Date: 2026-03-13  09:06:43
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:06:53 ; End-Date: 2026-03-13  09:09:06
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:09:10 ; End-Date: 2026-03-13  09:09:47
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:09:50 ; End-Date: 2026-03-13  09:10:29
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:10:37 ; End-Date: 2026-03-13  09:11:24
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:11:29 ; End-Date: 2026-03-13  09:12:08
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:12:11 ; End-Date: 2026-03-13  09:14:48
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:14:52 ; End-Date: 2026-03-13  09:16:57
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:17:01 ; End-Date: 2026-03-13  09:18:15
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:18:19 ; End-Date: 2026-03-13  09:18:57
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:19:00 ; End-Date: 2026-03-13  09:19:57
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:20:01 ; End-Date: 2026-03-13  09:21:26
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:21:30 ; End-Date: 2026-03-13  09:22:28
Commandline: /usr/bin/unattended-upgrade ; Start-Date: 2026-03-13  09:22:32 ; End-Date: 2026-03-13  09:22:55
```

Раздел 3

1-3. Зарегистрируйтесь на сайте github.com, если еще не сделали этого ранее

Установите параметры git в соответствии с вашими учетными данными:

git config --global user.name "имя пользователя github"

git config --global user.email "ваш e-mail, использованный при регистрации"

git config --global init.defaultBranch main

Установите непосредственно программу **git** и набор утилит для управления github из командной строки **gh**:

apt update

apt install git gh

```
root@eltex-practice2-pg2-v21:~# git config --global user.name "Artem06092005"
root@eltex-practice2-pg2-v21:~# git config --global user.email "chirkin.aleksandrowich@yandex.ru"
root@eltex-practice2-pg2-v21:~# git config --global init.defaultBranch main
root@eltex-practice2-pg2-v21:~#
```

4. Создайте ключ ssh:

ssh-keygen -t ed25519 -f ~/.ssh/название_ключа (имя пользователя github или имя пользователя для первой практики, или на ваш вкус)

ssh-keygen -t ed25519 -f ~/.ssh/github_key

5. Подключите этот комьютер к github:

   gh auth login

   ? What account do you want to log into? **GitHub.com**

   ? What is your preferred protocol for Git operations on this host? **SSH**

   ? Upload your SSH public key to your GitHub account?

   /root/.ssh/название_ключа.pub

   ? Title for your SSH key: eltex-practice2-pg**X**-v**Y**

   ? How would you like to authenticate GitHub CLI? **Login with a web browser**

```
root@eltex-practice2-pg2-v21:~# gh auth login
? What account do you want to log into? GitHub.com
? What is your preferred protocol for Git operations on this host? SSH
? Upload your SSH public key to your GitHub account? /root/.ssh/github_key.pub
? Title for your SSH key: eltex-practice2-pg2-v21
? How would you like to authenticate GitHub CLI? Login with a web browser
```

6. Укажите автоматическую авторизацию только для сайта github.com:

   gh auth setup-git --hostname github.com

   Убедиться, что все в порядке можно командой: gh auth status

```
root@eltex-practice2-pg2-v21:~# gh auth setup-git --hostname github.com
root@eltex-practice2-pg2-v21:~# gh auth status
github.com
  ✓ Logged in to github.com account Artem06092005 (/root/.config/gh/hosts.yml
  - Active account: true
  - Git operations protocol: ssh
  - Token: gho_****************************************
  - Token scopes: 'admin:public_key', 'gist', 'read:org', 'repo'
root@eltex-practice2-pg2-v21:~#
```

7. Создайте репозиторий на сервере github и наполняем его:

   gh repo create --public название_репозитория

```
root@eltex-practice2-pg2-v21:~# gh repo create --public eltex_modl
✓ Created repository Artem06092005/eltex_modl on GitHub
  https://github.com/Artem06092005/eltex_modl
root@eltex-practice2-pg2-v21:~#
```

```
root@eltex-practice2-pg2-v21:~# mkdir ~/eltex_modl && cd ~/eltex_modl
root@eltex-practice2-pg2-v21:~/eltex_modl#
<ex_modl# echo "Eltex модуль 1 практика" >> README.md
root@eltex-practice2-pg2-v21:~/eltex_modl# git init
Initialized empty Git repository in /root/eltex_modl/.git/
root@eltex-practice2-pg2-v21:~/eltex_modl# gid add README.md
Command 'gid' not found, but can be installed with:
apt install id-utils
root@eltex-practice2-pg2-v21:~/eltex_modl# git add README.md
root@eltex-practice2-pg2-v21:~/eltex_modl# git commit -m "first commit"
[main (root-commit) ca64a3c] first commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
root@eltex-practice2-pg2-v21:~/eltex_modl# git branch -M main
<ователя_github/название_репозитория.git
```

```
root@eltex-practice2-pg2-v21:~/eltex_mod1# git remote add origin https://github.com/Artem06092005/e>
root@eltex-practice2-pg2-v21:~/eltex_mod1# git status
On branch main
nothing to commit, working tree clean
root@eltex-practice2-pg2-v21:~/eltex_mod1#
root@eltex-practice2-pg2-v21:~/eltex_mod1# git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 265 bytes | 265.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Artem06092005/eltex_mod1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
root@eltex-practice2-pg2-v21:~/eltex_mod1#
```

```
eltex-pg2-v21@eltex-2026-spring:~$ scp task1.pdf root@172.16.9.221:/root/eltex_mod1/module1/task1
task1.pdf                                                     100%  522KB 116.2MB/s   00:00
eltex-pg2-v21@eltex-2026-spring:~$
```

```
root@eltex-practice2-pg2-v21:~/eltex_mod1# git add .
root@eltex-practice2-pg2-v21:~/eltex_mod1# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   module1/task1/task1.pdf

root@eltex-practice2-pg2-v21:~/eltex_mod1# git commit -m "1 отчёт"
[main e4a7a2d] 1 отчёт
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 module1/task1/task1.pdf
root@eltex-practice2-pg2-v21:~/eltex_mod1# git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 469.92 KiB | 36.15 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Artem06092005/eltex_mod1.git
   ca64a3c..e4a7a2d  main -> main
root@eltex-practice2-pg2-v21:~/eltex_mod1# ls
module1  README.md
root@eltex-practice2-pg2-v21:~/eltex_mod1#
```

8. Клонируйте ваш репозиторий:

   mkdir ~//eltex_mod1_clone && cd ~/eltex_mod1_clone

```
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone# git clone https://github.com/Artem06092005/eltex_mod1.git
Cloning into 'eltex_mod1'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 8 (delta 0), reused 8 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), 470.15 KiB | 1.28 MiB/s, done.
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone# ls
eltex_mod1
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone# rm -r eltex_mod1
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone#
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone#
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone# git clone https://github.com/Artem06092005/eltex_mod1.git .
Cloning into '.'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 8 (delta 0), reused 8 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), 470.15 KiB | 1005.00 KiB/s, done.
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone# ls
module1  README.md
root@eltex-practice2-pg2-v21:~/eltex_mod1_clone#
```

9. Внесите изменение в основном каталоге, не забыв про git add/commit/push, выполните в каталоге с клоном:
git fetch
git log origin/main
git diff main origin/main
git merge (для автоматического слияния используйте git pull вместо fetch+merge)

Вносим изменения:

```
root@eltex-practice2-pg2-v21:~/eltex_modl# echo "что-то новое" >> README.md
root@eltex-practice2-pg2-v21:~/eltex_modl# git add README.md
root@eltex-practice2-pg2-v21:~/eltex_modl# git commit -m "update Readme"
[main 8a74bf3] update Readme
 1 file changed, 1 insertion(+)
root@eltex-practice2-pg2-v21:~/eltex_modl# git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 348 bytes | 348.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Artem06092005/eltex_modl.git
   e4a7a2d..8a74bf3  main -> main
root@eltex-practice2-pg2-v21:~/eltex_modl#
```

git log origin/main :

```
    first commit
root@eltex-practice2-pg2-v21:~/eltex_modl_clone# git fetch
root@eltex-practice2-pg2-v21:~/eltex_modl_clone# git log origin/main
WARNING: terminal is not fully functional
Press RETURN to continue
commit 8a74bf3a950a8268e4658f641690677f801021cf (origin/main, origin/HEAD)
Author: Artem06092005 <chirkin.aleksandrowich@yandex.ru>
Date:   Mon Mar 16 13:08:11 2026 +0000

    update Readme

commit e4a7a2d553517095dca96f5660b2a013638b4c89 (HEAD -> main)
Author: Artem06092005 <chirkin.aleksandrowich@yandex.ru>
Date:   Mon Mar 16 12:51:38 2026 +0000

    1 отчёт

commit ca64a3cee627069fed0d1441b5840c57c184bb16
Author: Artem06092005 <chirkin.aleksandrowich@yandex.ru>
Date:   Mon Mar 16 11:56:42 2026 +0000

    first commit
```

git diff main origin/main :

```
root@eltex-practice2-pg2-v21:~/eltex_modl_clone# git diff main origin/main
WARNING: terminal is not fully functional
Press RETURN to continue
diff --git a/README.md b/README.md
index ebac08c..032a69d 100644
--- a/README.md
+++ b/README.md
@@ -1 +1,2 @@
 Eltex модуль 1 практика
+что-то новое
root@eltex-practice2-pg2-v21:~/eltex_modl_clone#
```

git pull :

```
root@eltex-practice2-pg2-v21:~/eltex_modl_clone# git pull
Updating e4a7a2d..8a74bf3
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
root@eltex-practice2-pg2-v21:~/eltex_modl_clone#
```